

C++ Memory Tracker CLI Tool

Task Name

C++ Memory Tracker CLI Tool

Description

Build a command-line tool that parses C++ source files to visualize pointer dependencies and dynamic memory allocations with color-coded output.

Your Claude Code Prompt

Act as an expert C++ developer and systems programmer. I need you to build a lightweight, standalone command-line prototyping tool in Python or Node.js that parses raw C++ source files to track dynamic memory management and pointer dependencies.

The tool needs to scan the target C++ code and identify:

Allocations: Where memory is initialized, such as `new`, `new[]`, `malloc`.

Dependencies and Passes: Where those pointers are assigned, passed into functions, or reassigned.

Deallocations: Where that memory is freed, such as `delete`, `delete[]`, `free`.

Output Format:

Print a clean, vertical, color-coded visual tree directly in the terminal using ANSI colors that maps the lifecycle of each pointer.

Use Green for allocations, such as `[ALLOCATED] ptr -> line 14`.

Use Yellow for transfers and dependencies, such as `%%% [PASS]`.

Use Red for deallocations, such as `%%% [FREED] -> line 45`.

If a pointer is allocated but never freed in the scanned scope, flag it at the bottom of its tree in Bold Red as a `[POTENTIAL LEAK]`.

Give me the complete, runnable script for this tool and a quick breakdown of how to execute it against a C++ file.